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International Mathematics Assessments for Schools

2024/2025 UPPER PRIMARY DIVISION FIRST ROUND PAPER

Time allowed：75 minutes

When your teacher gives the signal, begin working on the problems.

INSTRUCTION AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. No calculators, slide rules, log tables, math stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 20 multiple-choice questions, each with 5 choices. Choose the most reasonable answer. The last 5 questions require whole number answers between 000 and 999 inclusive. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a mathematics assessment, not a test; do not expect to answer all questions.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are filled in. It is your responsibility that the Answer Sheet is correctly coded.

THE ANSWER SHEET

1. Use only pencils.
2. Record your answers on the reverse side of the Answer Sheet (not on the question paper) by FULLY filling in the circles which correspond to your choices.
3. Your Answer Sheet will be read by a machine. The machine will see all markings even if they are in the wrong places. So please be careful not to doodle or write anything extra on the Answer Sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

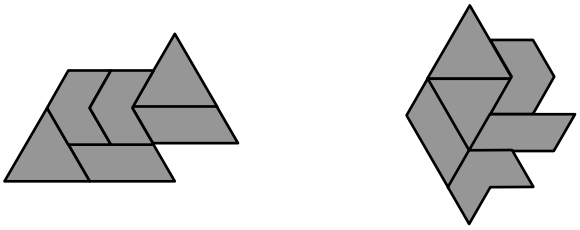
INTEGRITY OF THE COMPETITION

The IMAS reserves the right to re-examine students before deciding whether to grant official status to their scores.

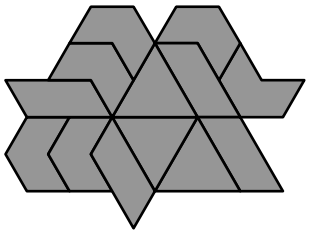
2024 / 2025 UPPER PRIMARY DIVISION FIRST ROUND PAPER

Questions 1-10, 3 marks each

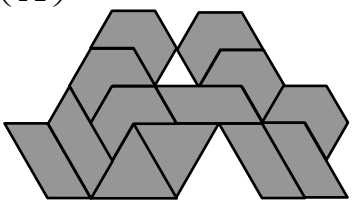
1. The diagram below shows two different patterns that were formed by using the same set of shapes, in the same numbers, without overlaps. (Note: If one shape matches another shape after rotations or reflections, then they are viewed as the same shape.)



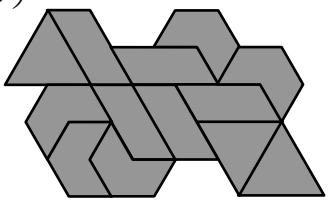
Which pattern from options (A) - (E), uses the same set of shapes, in the same numbers with the following diagram?



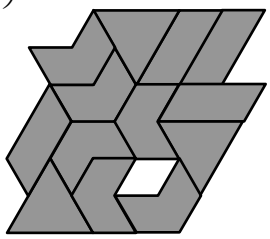
（A）



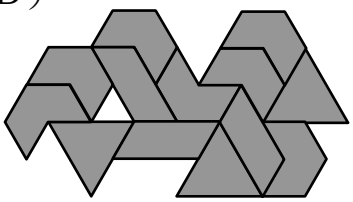
（B）



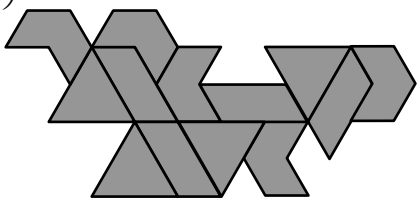
（C）



（D）



（E）



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2. A tour company offers two types of bus: a 26-seater or a 39-seater passenger bus. A group consisting of 212 tourists wants to use the buses provided by this company. If the company wants to optimize the selection of the buses so that the total number of empty seats is as small as possible, what is the smallest possible number of empty seats?

（A）3　　（B）5　　（C）8　　（D）9　　（E）12

---

3. A product lost 60% of its value this year. What must be the percentage increase of its value next year to be able to regain its original value?

（A）50%　　（B）60%　　（C）120%　　（D）150%　　（E）200%

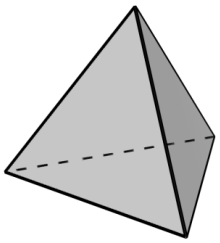
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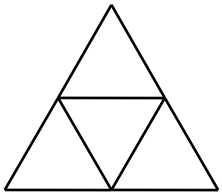
4. A recurring decimal is a decimal representation of a number whose digits are periodic in nature, meaning that, after some places, the same sequence of digits repeats continuously. For example, $\frac{1234}{9999} = 0.12341234... = 0.\dot{1}2\dot{3}4\dot{}$.

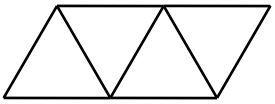
   It is known that the recurring decimal of $\frac{1}{17}$ is 16 digits long. What is the sum of all the digits of the recurring decimal of $\frac{1}{17}$?

   （A）71　　（B）72　　（C）73　　（D）74　　（E）75

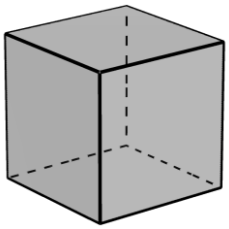
5. A polyhedra is a three-dimensional solid with flat faces, straight edges and vertices. Models of polyhedra are made from flat diagrams drawn on a board. On the diagram the faces are all connected, and one makes a model of it by folding the board along the lines of the diagram. A regular tetrahedron has two different diagrams as shown below:



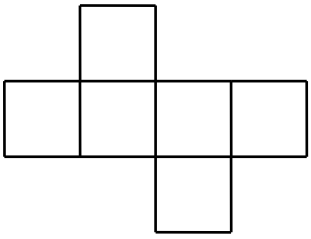




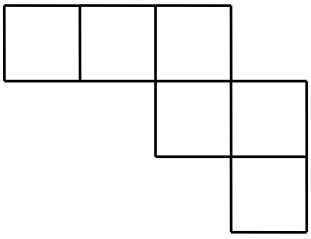
Which diagram from options (A) - (E) is **NOT** a flat diagram of a cube?



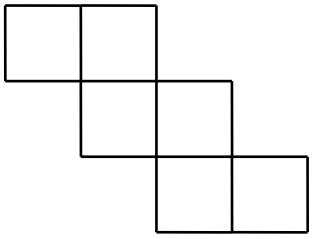
（A）



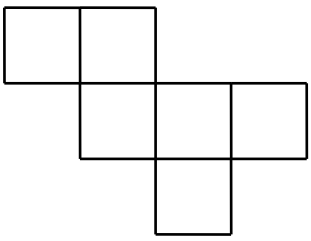
（B）



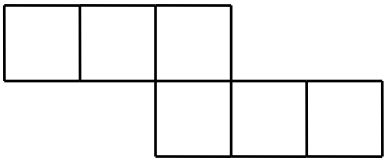
（C）



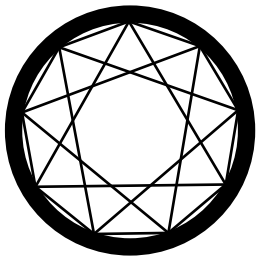
（D）

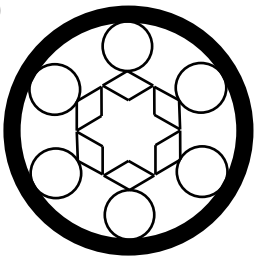


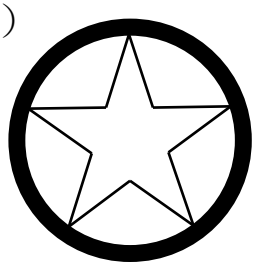
（E）

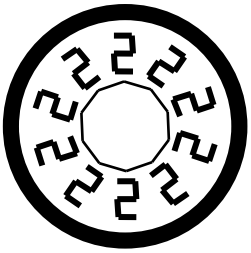


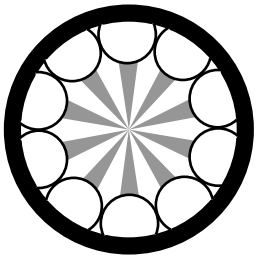
6. Which shape among options (A) - (E) has EXACTLY 10 axes of symmetry?

（A） 

（B） 

（C） 

（D） 

（E） 

---

7. A triangle has an area of 120 cm^2. If its base is increased by 50% and its height is decreased by 25%, then what will be the area, in cm^2, of the new triangle?

（A）75　（B）90　（C）120　（D）135　（E）150

---

8. Five students are assigned to clean a classroom, where each student has the same efficiency work rate. One hour after they started, two of the students left for a scheduled singing practice. After two more hours, one of the remaining students left for a scheduled basketball practice. If the remaining two students completed the rest of the job in 1 hour, how long would it take for them to finish cleaning if all five students have stayed from the beginning?

（A）2 hours 36 minutes　（B）2 hours 44 minutes　（C）2 hours 52 minutes
（D）3 hours　（E）3 hours 8 minutes

---

9. A machine transmits messages that consist only of the letters A, B, C, D and E. Before the machine sends a message, A is encoded as 11, B as 10, C as 011, D as 010 and E as 00. For example, the message CABDE is encoded as 011111001000. If this machine sent 100111010111100010, what was the message?

（A）BCBBBADE　（B）BCBBAABD　（C）BCBBAEAD
（D）BCBBAAED　（E）BCAABBED

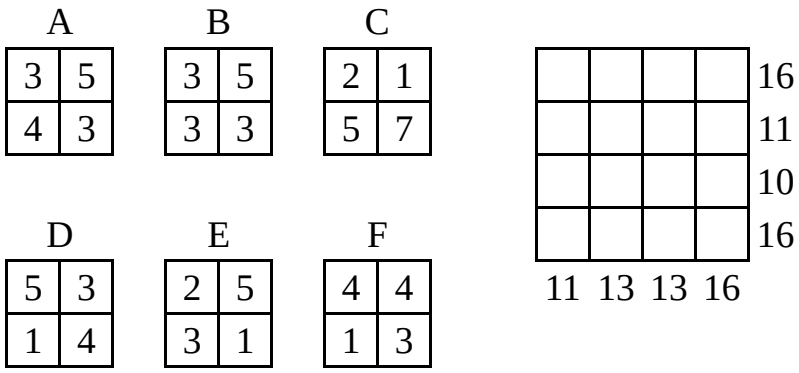
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10. The sum of the ages of Ann, Bob, Chris and Dan is 50. For each pair of them, the sum of their ages is at least 21. What is the minimum age that Ann can have? (Note: all ages are expressed as integer values)

（A）6　（B）7　（C）9　（D）10　（E）12

Questions 11-20, 4 marks each

11. Select four pieces of 2×2 grid (which are labelled as A, B, C, D, E and F) and place them in the 4×4 grid such that the sum of numbers in each row equals the number written in the right-most side of each corresponding row and the sum of numbers in each column equals the number written in the bottom-most of each corresponding column.
(Note: it is NOT allowed to rotate or reflect any of the 2×2 grids.)



Which of the following options is a correct solution?

(A)

D	B
C	E

(B)

B	D
C	F

(C)

D	B
E	C

(D)

E	C
D	B

(E)

A	F
B	D

---

12. Select six different digits from 0, 1, 2, 3, 4, 5, 7, 8 and 9, and then use these digits to create two positive three-digit numbers. What is the smallest possible positive difference (subtract the smaller number from the larger number) for these two three-digit numbers?

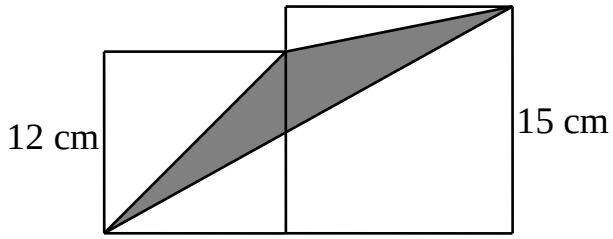
(A) 3 (B) 4 (C) 5 (D) 12 (E) 14

---

13. Let N be a positive integer so that $N\times N = 44\ldots488\ldots89$ (2025 fours, 2024 eights). What is the value of N?

(A) $66\ldots63$ (2023 sixes) (B) $66\ldots67$ (2023 sixes) (C) $66\ldots63$ (2024 sixes) (D) $66\ldots67$ (2024 sixes) (E) $66\ldots63$ (2025 sixes)

14. Two squares, with sides 12 cm and 15 cm, are placed side by side, without overlapping, as shown in the diagram below. What is the area, in cm^2, of the shaded triangle?



（A）75　（B）72　（C）69　（D）66　（E）63

15. The ladybug which is located on the left of the diagram wants to meet up with her friend located on the right. To get there, she will have to cross a field which consists of four kinds of flowers. Each kind of flower represents a different direction --- either up, down, left or right. The black rectangles are deep pits that must be avoided.



It is known that there is a single path so that the ladybug can meet her friend. From options (A) - (E), which direction represented by the flower is correct?

（A）Rose represents up　（B）Rose represents left
（C）Rose represents right　（D）Orchid represents left
（E）Lily represents left

16. A shipping company offers three different sizes of shipping boxes namely:
    - Size A: 12 m × 9 m × 6 m
    - Size B: 15 m × 12 m × 8 m
    - Size C: 18 m × 10 m × 9 m

    Suppose that the r pieces of Size A have the same total volume as s pieces of Size B, which also have the same total volume as t pieces of Size C. What is the minimum value of $r+s+t$?

（A）33　（B）34　（C）35　（D）36　（E）37

17. How many three-digit numbers, which are composed of three distinct digits do not contain the digit “0”, are divisible by 6 or 15?

（A）100　（B）96　（C）94　（D）92　（E）90

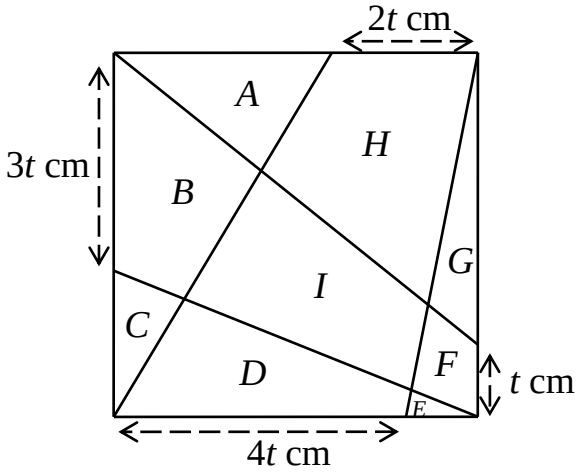
18. How many numbers between 1 to 100, inclusive, have exactly six positive divisors?

（A）16　　（B）15　　（C）14　　（D）13　　（E）12

19. Select an even number of distinct integers from the set $\{1, 2, \ldots, n\}$ and group them into pairs. For each pair, find the positive difference of the two values (subtract the smaller number from the larger number), and then take the product of all such differences. If this product is equal to 2025, what is the least possible value of n?

（A）9　　（B）10　　（C）12　　（D）14　　（E）15

20. A square is partitioned by segments into 9 regions so that the area of region I is equal to the total area of regions A, C, E and G, as shown in the diagram below, where the lengths of the indicated segments are shown. If the area of the square is 350 cm^2, what is the area, in cm^2, of the triangle formed by regions C, D and E?

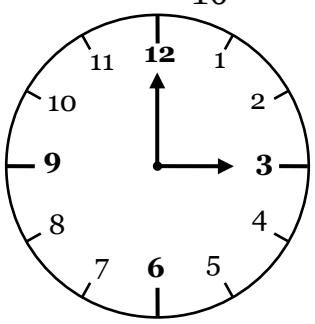


（A）35　　（B）50　　（C）70　　（D）135　　（E）140

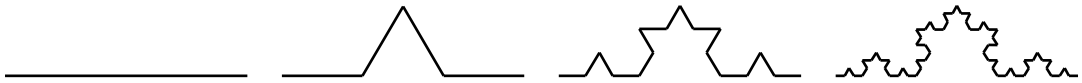
Questions 21-25, 6 marks each

21. Consider an analog clock with hour and minute hands, where the current time is 3 o'clock. In t minutes, the hour and minute hands will be equidistant from the 12 o'clock direction in terms of angles, but the two hands are not overlapping.

What is the smallest possible value of $\dfrac{143t}{10}$?



22. A fractal is a pattern that repeats forever, and every part of it, regardless of how zoomed in or zoomed out you look at the picture, will pretty much look similar to the whole image. Koch's curve is a fractal that has infinite length but no area, generated by the following process, as shown in the diagram below:
    - At iteration 0, draw a horizontal line segment that has length 1.
    - At each of the iterations 1, 2, …:
      Each line segment from the previous iteration is divided into three equal segments, and the middle line segment is replaced by the two sides of an equilateral triangle of the same length as the segment being removed.



iteration 0 iteration 1 iteration 2 iteration 3

Let T_n denote the length of the curve obtained at iteration *n*. What is the least integer *n* such that $T_n > 5$?

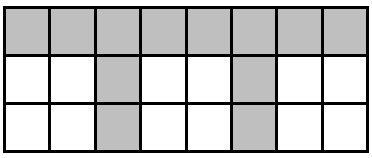
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23. Benny has 9 cards, where each card contains one digit written from 1 to 9. The digit written in different cards are different. He takes three cards so that the average of the three cards is equal to the average the other six cards. In how many ways can he do this?



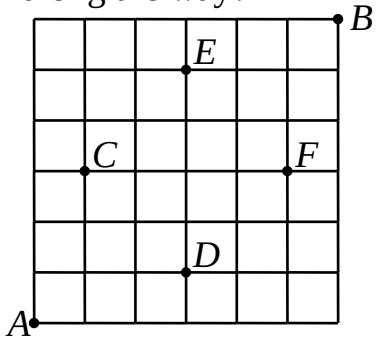
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24. The diagram below shows a 3×8 board with 16 unit cells coloured grey. In how many ways can one fill the numbers 1, 2, …, 12 into the grey cells, where each cell can be filled with exactly one number, such that the numbers on the first row are increasing from left to right and the numbers on the third column and the sixth column are increasing from top to bottom?



---

25. In the following diagram, how many paths are there to travel from point *A* to point *B* by going either right and up, while passing through exactly two of the four points *C*, *D*, *E* and *F* along the way?



***